

02 Conversion

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6:08 PM

In this class, we denote conversion with the letter X

The definition of conversion is $X = \frac{\text{amount reacted}}{\text{amount fed (flow reactor) or present initially (batch reactor)}}$

Mathematically, $X = \frac{F_{A0} - F_A}{F_{A0}}$ in a flow reactor

And $X = \frac{N_{A0} - N_A}{N_{A0}}$ in a batch reactor

For the special case where there is only 1 reaction and the mole balances are only influenced by the reaction (and not by moles being added or removed by other means such as diffusion through a membrane), it is useful to *solve* the mole balance in terms of X , as this reduces a multidimensional problem to "1 equation and 1 unknown".

Let's convert the mole balances for the 3 common reactors so that they are in terms of X instead of F_A or N_A

CSTR:

$$V = \frac{F_{A0} - F_A}{-r_A} \text{ and } X = \frac{F_{A0} - F_A}{F_{A0}}.$$

Step 1: Write F_A in terms of X :

$$X = \frac{F_{A0} - F_A}{F_{A0}}, \text{ therefore } F_A = F_{A0}(1-X)$$

Step 2: Substitute into the mole balance:

$$V = \frac{F_{A0} - F_A}{-r_A} = \frac{F_{A0} - F_{A0}(1-X)}{-r_A} = \frac{F_{A0}X}{-r_A}$$

Batch reactor:

$$\frac{dN_A}{dt} = r_A V \text{ and } X = \frac{N_{A0} - N_A}{N_{A0}}$$

Write N_A in terms of X is $X = \frac{N_{A0} - N_A}{N_{A0}}, \therefore N_A = N_{A0}(1-X)$

Hence the mole balance in terms of X is:

$$\frac{dN_A}{dt} = \frac{d}{dt} (N_{A0}(1-X)) = \frac{d}{dt} (N_{A0} - N_{A0}X) = \frac{dN_{A0}}{dt}$$

$$\frac{dN_{A0}}{dt} = 0 \quad \therefore \frac{d(N_{A0}X)}{dt} = N_{A0} \frac{dX}{dt}$$

N_{A0} is constant, so $\frac{dN_{A0}}{dt} = 0$

PFR: Work

$$\frac{dF_A}{dV} = r_A \text{ and } X = \frac{F_{A0} - F_A}{F_{A0}}, \quad F_A = F_{A0}(1 - X)$$

$$\frac{d}{dV} (F_{A0}(1 - X)) = -F_{A0} \frac{dX}{dV} \quad \text{Therefore, } \frac{dX}{dV} = \frac{-r_A}{F_{A0}}$$

F_A in terms of X is $F_A = F_{A0}(1 - X)$. Prove that the mole balance in terms of X is: $F_{A0} \frac{dX}{dV} = -r_A$

$$\frac{dF_A}{dV} = \frac{d}{dV} (F_{A0}(1 - X)) = \frac{d}{dV} (F_{A0} - F_{A0}X)$$

$$\frac{d}{dV} F_{A0} - \frac{d}{dV} (F_{A0}X) = -F_{A0} \frac{dX}{dV}$$

$$\frac{dF_A}{dV} = -F_{A0} \frac{dX}{dV} \quad \therefore \quad F_{A0} \frac{dX}{dV} = -r_A$$